

Building Research & Information

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/rbri20

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To cite this article: Sathishkumar V E , Changsun Shin & Yongyun Cho (2020): Efficient energy consumption prediction model for a data analytic-enabled industry building in a smart city, Building Research & Information, DOI: 10.1080/09613218.2020.1809983

To link to this article: https://doi.org/10.1080/09613218.2020.1809983

Published online: 17 Sep 2020.

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BRI BUILDING RESEARCH & INFORMATION
https://doi.org/10.1080/09613218.2020.1809983

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Efficient energy consumption prediction model for a data analytic-enabled industry building in a smart city

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ABSTRACT

The fast development of urban advancement in the past decade requires reasonable and realistic solutions for transport, building infrastructure, natural conditions, and personal satisfaction in smart cities. This paper presents and explores predictive energy consumption models based on data-mining techniques for a smart small-scale steel industry in South Korea. Energy consumption data is collected using IoT based systems and used for prediction. Data used include the lagging and leading current reactive power, the lagging and leading current power factor, carbon dioxide emissions, and load types. Five statistical algorithms are used for energy consumption prediction:(a) General linear regression, (b) Classification and regression trees, (c) Support vector machine with a radial basis kernel, (d) K nearest neighbours, (e) CUBIST. Root mean squared error, Mean absolute error and Coefficient of variation are used to measure the prediction efficiency of the models. The results show that CUBIST model provides best results with lower error values and this model can be used for the development of energy efficient structural design which helps to optimize the energy consumption and policy making in smart cities.

ARTICLE HISTORY
Received 21 May 2020
Accepted 9 August 2020

KEYWORDS
Data mining; energy consumption; feature ranking; data analysis

Introduction

Several Governments of metropolises are adopting the smart cities concept. For which they begin to collect big datasets for obtaining valuable information. Using this information, governments intend to develop the living standards and sustainability for the inhabitants. Optimizing different energy resource consumption and reducing the cost are vital for rising the life quality and comfort of people. Considering the cost reduction, it can help to increase the efficiency in areas like health-care, education, security, transport, and emergency services. Concerning this situation, there is a widespread use of smart grid technologies for massive data storage. For instance, in public institutions, water or electricity energy consumption is monitored continuously. However, the high computational cost of mining gigabytes of data lead to the extinction of conventional tools and storing valuable information extraction techniques. Eventually, mining data became easier with the introduction of data-mining tools. But, these techniques need to be improved in the area of management, processing and finding of useful information and knowledge for organizations.

Recently, Smart city infrastructure tries to address or limit the problems created by quick urbanization and financial advancements, for example, electricity production, removal and management of waste, and modes of transportation, through the best execution and utilization of resources. In the past decade, a few sorts of researches incorporate various orders of smart city improvement zones, where the disadvantage of the energy utilization progressive system depends on the intelligent frameworks which needs data and forecasting methodologies to make such smart system. In General, Smart Cities have various and enormous energy requirements for managing all the components in a smart city which present new advances in an organized manner by exploiting all these energy sources, in an ideal way. So an efficient prediction model for managing energy in a smart city is required. Figure 1 presents various components of a smart city.

The advancement in Industrialization increases the energy demand, for which it is considered as an important part of national policy. In addition, the energy utilization expands with the economic growth and human advancements. Uncontrolled energy consumption (for example, over consumption, weak systems, and waste energy) serves as a reason for this phenomenon. Due to the steady rise in demand, energy is deemed one of the most significant and scarce resources. Energy saving is sought not only for advancing a green environment for

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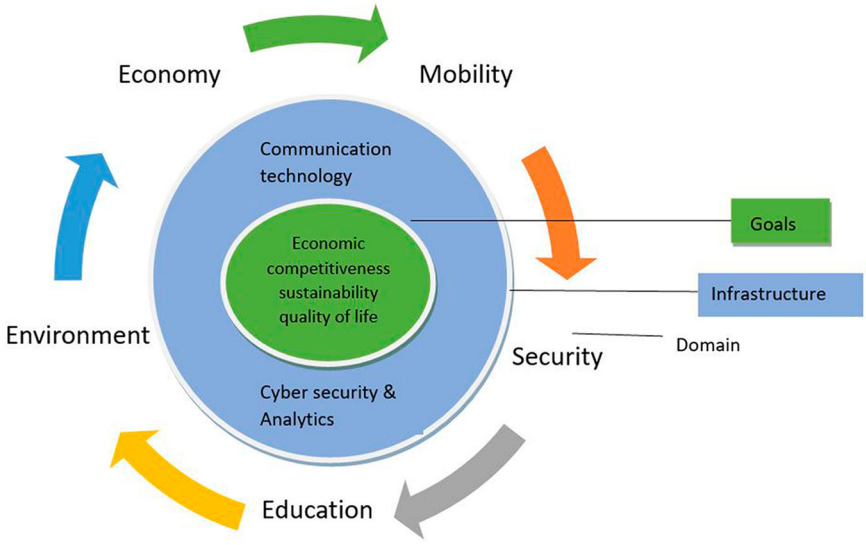


Figure 1. Components of smart city.

future efficiency, but also for the benefit of domestic consumers and energy production companies. Rapid economic growth adds to the emerging calls for electrical power worldwide. In the meantime, electricity is deemed to be among the most fundamental elements pushing economic growth and is held to be vital in our domestic lives (Xiao et al., 2016). Consequently, for a nation or a region, forecasting electrical energy usage has become pressing and crucial (Wang et al., 2018). So, proper planning is required for building efficient energy management systems.

One of the main sectors that needs energy stabilization is industrial sector. Since the 1990s, the manufacturing industry in South Korea has evolved at an elevated speed and has become a primary leading power in the South Korean economy's ongoing fast growth. Primary energy usage increased in the 1990s at a yearly pace of 7.5%, greater than the annual financial development rate of 6.5% during the same time. This is due to the sharp progression in energy-intensive industries, including petrochemicals. The steep rise in industrial electricity consumption has facilitated the hike in loss of energy conversion, further undermining energy intensity. The rise in energy-industry production after 2009 considerably buffered the nation in its resistance to the global economic crisis, but it adversely affected the gross energy efficiency of the nation (Kim et al., 2001). Several uncertain factors like industrial structure, the level of technology, cost of energy, economic scale and national policy have an impact on industrial energy consumption.

Life is made simple with the accomplishments of third scientific and technological revolution. It has enhanced the innovative technology and systemic change in conventional industries. Manufacturing industry is a prominent sector for both nation and country and serves as a key financial level index. Despite having advanced manufacturing sectors, several developed countries seek to explore new possibilities and attempts to redesign the manufacturing industries for acquiring an unconquerable position in the presence of technological growth and modernization. Germany is a typical example for this trend. Relying on the smart growth, its fourth industrial revolution (a.k.a. Industry 4.0) focuses on output effectiveness, material use and energy consumption (Xiao et al., 2016). Despite the usefulness of the findings in literature, it has its own drawbacks which includes discussion that are comparatively macroscopic, and it lacks empirical industrial research. This article focuses on predicting the South Korean steel industry's energy consumption.

Related work

Recently, using data-mining techniques, many scientists have examined the energy consumption prediction problem. For modelling the energy consumption patterns, several statistical and Artificial Intelligence (AI) methods are developed. After establishing the model, machine learning algorithms are very useful, and convenient for a normal operator to use; they are now more popular in many applications (Simeone, 2018).

Many surveys ascertain the outstanding importance of enhancing the efficiency of energy for financial growth (Arbex & Perobelli, 2010; Carley et al., 2011). A study based on a recommendation system was carried out for service suggestions in a cloud computing environment. This recommendation system can also be used in energy management systems. Nasreen and Anwar's (Nasreen &

Anwar, 2014) study of the relation between Asia's economic growth, trade and energy reveals the means by which economic growth and open trade affects energy use in a constructive manner. Several researchers affirm a one-way economic-energy relation (Kraft & Kraft, 1978; Lee, 2005; Tsani, 2010). On the other hand, some render evidence for a two-way relationship between them. To build enterprise irrespective of the type, Knowledge mining is always important (Tajuddin et al., 2019). Some expert studies analysing the correlation between population and green economy show that green economy benefits employment in majority of the countries (Lombardi & Schwabe, 2017).

To forecast electricity requirements, researchers typically use designs such as multiple regression (Candanedo et al., 2017), support vector machines (Cherkassky & Ma, 2004), support vector machines with principal component analysis (Jinhu et al., 2010), time series techniques (Alsharif et al., 2019), gradient boosted machines (Candanedo et al., 2017), artificial neural networks (Le & Jing, 2015), and deep learning methods (Mocanu et al., 2016). The parameters which the models consider are the period of day, outside temperature, the month, whether it is a weekday/weekend or holiday, energy usage of the previous day, the rainfall index, global solar radiation, wind speed and occupancy (Bacher et al., 2013; Sandels et al., 2015). Since all patterns of energy consumption are generally linked to certain models of time cycles, some of the characteristics obtained from date and time stamps are favourable in designing predictive models. Classification and regression tree (CART) give a precise forecast of building energy demand with minimum error (Yu et al., 2010). This technique classifies and foresees categorical factors with an interpretable flowchart, like tree structures, allowing consumers to obtain helpful data quickly.

Statistical procedures have a limited performance as the energy demand patterns are unpredictable. Further, several predictions using AI techniques are explored. Using SVM with climate data and energy utilization, Dong et al forecasted the demand for building energy (Dong et al., 2005). Having the upcoming temperature from the temperature to date, using a feedforward neural network Gonzalez and Zamarreno calculated the demand from the difference between them (Gonzalez & Zamarreno, 2005). Ekici and Aksoy excluded the climate conditions and used the building property to estimate the building's energy needs (Ekici & Aksoy, 2009). Whereas, Li et al. used an SVM with the transfer coefficient of the building to predict the annual demand. These investigations resulted in the development of models that predicts the right values which correspond to the independent variables. In an attempt to arrive at a solution to this issue, Xuemei et al. used Fuzzy C-means clustering and predicted the demand with a fuzzy SVM (Xuemei et al., 2010).

Considering the diverse production and usage of Classification and Regression Trees as foundation models, Random Forest can be asserted as an ensemble of numerous Classification and Regression Trees. Random Forest model is used in many research fields. For instance, in identification issues using RF (Díaz-Uriarte & De Andres, 2006), a novel procedure for gene selection is suggested. Cutler et al. used RF in the ecology field for the purpose of differentiating various kinds of plant species (Cutler et al., 2007). In Remote sensing field Rodriguez-Galiano et al. used RF to differentiate varying land cover (Rodriguez-Galiano et al., 2012). On the other hand, Sun et al. employed RF for the purpose of forecasting solar radiation from different places (Sun et al., 2016). These study results show that RF is efficient enough in handling the issues of classification and regression. In addition, RF is compared to other prominent methods like linear discriminant analysis, decision trees, and support vector machine in earlier research programs. Study findings indicate that RF outperforms these challengers in fixing the study issues, showing their ability as a promising tool to solve the issue of building energy prediction.

K-nearest neighbours (KNN) is used to measure the building energy consumption for the next day (Fan et al., 2014). Here, the performance of the KNN model is stronger than that of the Auto Regressive Integrated Moving Average (ARIMA) model and the Neural Networks. Zha et al. presented a grey forecast model to address the predictive problem for the manufacturing industry in China (Zha et al., 2009). This work extracts and constructs a homologous grey prediction model based on the gray system theory, with single variable and single first-order equation for the energy consumption of Chinese manufacturing. Zhu et al. presented a logistic regression and grey model for assessing Tijanjin industry's energy consumption and provided constructive suggestions for future energy savings and carbon dioxide (tCO2) emissions control (Guerreiro & Belo, 2009). This model resolves parameter inaccuracy issues (between parameters) and works better than other grey prediction models. An efficient prediction strategy is one of the major issues in all computing environments (Guerreiro & Belo, 2009). Previous works agree that all machine learning models operate adequately to predict energy consumption trends. By considering the existing literature works, the Cubist algorithm has never been used in energy consumption research area. Therefore, in this analysis the efficiency of Cubist algorithm is used to estimate energy consumption. The Cubist

algorithm's strength is compared with the top-performing algorithms in prediction studies of energy consumption. This research is to comprehend the relation between industrial energy consumption with varying predictors. As well, we explore the efficiency of various machine learning models (GLR, CART, KNN, RF, and the CUBIST) in energy consumption prediction for the industry.

Methodology

General linear regression

General linear regression (GLR), a varied generalization of linear regression, reveals that the datasets have an arbitrary distribution model. Based on its distribution model, the connection between x (predictor factors) and y (reaction vector) is built using a connection feature. The design $(X–Y)$ is depicted in the following equation:

$$g(E(y)) = x \times \beta + O > y \sim F \tag{1}$$

Here $g(\bullet)$ is the link function, the predictor is x, the response variable is y, the offset variable is O and the regression coefficient is β.

GLR utilizes the Newton–Raphson approach to get a continuous estimate so that $g(E(y))$ reaches $(x\beta + O)$. The ultimate proximal equation is formulated as the relational expression $(X–Y)$. Extra parameters in GLR boost model flexibility. On the other hand, GLR has a vast application obtaining a reasonable model of connection, which is in opposition with that of linear regression.

Classification and regression tree

A decision tree-based technique Classification and regression trees (CART) could be used for numerical or categorical, is for building a regression or classification tree according to the form of target variable. While considering CART model which is rule based, the data is divided into subsets of two, so that fresh data subset has greater purity than the preceding subset. Reaching the criteria for homogeneity needs a splitting process in recursive manner. A CART based model is versatile enough to weigh misclassification expenses and indicate the prior distribution of likelihood in a problem of classification. Decision tree techniques in terms of their logic rules are considerably better than other techniques of modelling (Clementine, 2007).

It is possible to use three impurity measurements to detect CART models splits for a given target field. In addition, the target field is usually symbolized using Gini, whereas the lowest squared deviation technique is used to continuously pick continuous targets without specifying the choice. Formulas (2) and (3) in a CART define the Gini coefficient, $g(t)$, at node t:

$$g(t) = \sum_{j \neq i} P(j|t)P(i|t) \tag{2}$$

$$P(j|t) = \frac{P(\dot{j}, t)}{p(t)}; P(j, t) = \frac{\pi(j)N_j(t)}{N_j}; \text{and } P(t) = \Sigma P(j, t) \tag{3}$$

Here i and j are dependent field categories, prior probability value of category j is $\pi(j)$, $Nj(t)$ is the number of records in category j of node t and Nj is the number of records in category j of the root node Notably, when using the Gini index to quantify the change after splitting during tree growth, only node records and the root node with valid split-predictor values are used to calculate $Nj(t)$ and Nj, respectively.

K-nearest neighbours

KNN is a non-parametric learning algorithm used for classification and/or regression. In both cases the input includes the K closest training examples in the feature space. The yield varies according to whether KNN is used in the classification or regression.

- For the classification, KNN product is a member of the class. An object is categorized by a relative majority of its neighbours that assigns it among its closest K neighbours to the most prevalent category (K is a positive number).
- In the case of KNN regression the output is the property value for the object. Finding the average of the K nearest neighbour values gives the yield value.

As the explicit mapping relation between input and output is not learnt, it is considered to be non-parametric. To generate the final scores of the test data, the proximity of neighbouring input and the corresponding output in the training data is used. Before scoring, we calculate the parameter K that specifies the number of neighbouring observations considered. KNN is a simple learning algorithm and its efficiency is still considered optimal in practical applications.

Support vector machines

The support vector machine (SVM), found in 1996, aims classifying data in the finest hyperplane (Schiilkop et al., 1995). The data is transformed and moved to a high-

dimensional feature space where data can be separated linearly using kernel functions to distinguish two groups that cannot be separated linearly. Likewise, the SVM carries out non-linear mapping of data with kernel functions in this new high-dimensional feature space, then continues to provide linear regression. The SVM regression function is defined in the following equation:

$$f(x) = \sum_{i=1}^{m} (\alpha_i^* - \alpha_i) k(x_i, x) + b \quad (4)$$

The value of $k(x_i, x)$ is equal to the inner product of two vectors, x_i and x_j, in the feature space, $\phi(x_i)$ and $\phi(x_j)$, that is, $K(x_i,x_j) = \phi(x_i)\cdot\phi(x_j)$. All necessary computations can be done directly in the input space, without having to compute the map, $\phi(x)$, using kernels. Some of the popular kernel functions are: (1) linear kernel, $K(x_i,x_j) = x_i\cdot x_j$, (2) polynomial kernel, $K(x_i,x_j) = (x_i\cdot x_j + 1)^d$, (3) the radial basis function (RBF)kernel,- $k(x_i, \gamma_i) = \exp(-\gamma x_i - x_j^2)$, $\gamma > 0$, where d and γ are the kernel parameters. Among the SVM kernels, Radial Basis Kernel has shown best performance in many of the studies and was considered in this study.

Random forest

In 2001, Breiman developed Random Forests (RF) for classification and regression issues and acquired significance in appropriately modelling the input and output relation due to its robustness and flexibility (Breiman, 2001). RF adds to bagging an additional randomness layer. Besides constructing every tree using a distinct data bootstrap sample, random forests modify how the trees are designed for classification and regression problems. That node is divided into standard trees which use the perfect divide among all factors. Every node at that node is spitted into a random forest that uses the best a randomly selected subset of predictors.

The Steps involved in random forest algorithm is as follows.

Step 1: Draw tree boot strap samples from the original information.

Step 2: Develop an unpruned classification or regression tree with the defined change for each of the bootstrap samples; Instead of choosing the best split between all predictors at each node, randomly sample mtry predictors to choose the best split among those variables. (Bagging can be viewed as a unique case of random forests in mtry = p, number of predictors.)

Step 3: Forecast fresh data by aggregating *ntree* tree predictions, meaning majority votes to assign the average for regression.

CUBIST

Cubist model is based on Quinlan M5 tree. A set of laws '*if-after-after*' is generated through the Cubist method. Each of the generated rules has a linear multivariate model associated with it. The corresponding model is used to evaluate the predicted value, until the covariate set meets rule requirements (VE & Cho, 2020). The principal benefit of the Cubist approach is that the weights are more consistent through the introduction of several teaching boards and neighbourhoods. Thus, the Cubist model is established with a series of trees. The number of neighbours is used to change rules. The Cubistic approach uses a linear combination of the two models. The general view of the Cubist regression model is defined as follows: During a tree's growth, many branches and leaves are grown. It is possible to take a set of '*if-after-after*' rules as branches, while the terminal leaves can be viewed as an associated linear multivariable model. To calculate the predictive value, the related models will be applied to multiple covariates that correspond to the condition of a law. The Cubist model incorporates boosting by sequentially developing a number of trees with adjusted weights, with training committees (usually larger than one) similar to the 'boosting' process. The number of neighbours, in the Cubist style, is used to change the rule-based prediction.

The two linear models in the Cubist model are represented as in the Equation (5):

$$\zeta par = (1 - a)*\zeta(p) + a*\zeta(c) \quad (5)$$

Here, $\zeta(c)$ is the current model prediction and $\zeta(p)$ is a relative model prediction above it in the list. To make it easy to grasp the Cubist paradigm, it is called a process composed of several rules in which each rule is connected in the corresponding multivariate linear style. The corresponding model is then assigned to determine the correct expected value for the state of the law. Cubist paradigm is good and beneficial for beginners with bare knowledge in statistical learning or machine learning. The Cubist paradigm appears to be a successful regression approach and can be applied to a range of problems not yet used in the field of energy consumption prediction. Thus, it is used in this study. All the variables and their corresponding explanations are summarized in Table 1

Table 1. Notation table.

Notation	Description
$g(\bullet)$	Link function
x	Predictor
y	Response variable
O	Offset variable
β	Regression coefficient
i and j	Dependent field categories
$Nj(t)$	Number of records in category j of node
Nj	Number of records in category j of the root node
$\zeta(c)$	Current model prediction
$\zeta(p)$	Relative model prediction
y_i	Actual measurement (energy consumption),
$\hat{y}_i$	Predicted value from the regression algorithms
n	Number of measurements.
$\bar{y}_{data}$	Average energy consumption

Table 2. Load type and timings.

Load type	June–August	March–May, September–October	November–February
Light load	23:00–09:00	23:00–09:00	23:00–09:00
Medium load	09:00–10:00	09:00–10:00	09:00–10:00
	12:00–01:00	12:00–01:00	12:00–07:00
	17:00–23:00	17:00–23:00	20:00–22:00
Maximum load	10:00–12:00	10:00–12:00	10:00–12:00
	01:00–17:00	01:00–17:00	17:00–20:00
			22:00–23:00

Recorded energy data and description

This section describes the energy data used in this study. This work uses the energy consumption data for every 15 min, which in turn are used for building the prediction models using machine learning algorithms: (a) GLR, (b) SVM, (c) CART, (d) KNN, (e) RF, and (f) CUBIST algorithm. The information gathered is from the DAEWOO Steel Co. Ltd in Gwangyang, South Korea. It produces several types of coils, steel plates, and iron plates. The information on electricity consumption is held in a cloud-based system. The information on energy consumption of the industry is stored on the website of the Korea Electric Power Corporation (pccs.kepco.go.kr), and the perspectives on daily, monthly, and annual data are calculated and shown.

Figure 2 presents the overall flow of the proposed design. The steel industry machine equipment energy consumption is collected using smart meter where additional information about the energy consumption were collected and stored in cloud-based system. Data analytic algorithms were used to forecast the demand which in turn used for effective policy making of energy consumption and optimization.

This assessment focuses on the energy data (in Kwh) recorded for the industry every 15 min. The reporting interval of 15 min was selected to track rapid energy consumption variations. All data analysis was performed in R (Team, 2014). The data time period is 365 days (12 months). Table 2 provides the load type and timing of each month. Table 2 depicts the load timings shift from November to February. Figure 3 is the energy consumption picture for the whole period, and we can see from the figure that the energy consumption pattern shows high variability. Figure 4 displays the histogram and box plot table. The data distribution displays a lengthy tail. The median in the box plot is represented inside the blue rectangle with a thick black line and has a value of 5 kWh. The bottom whisker is 3 kWh whereas the top whisker is 125 kWh. The outliers are branded with circles over the upper whisker.

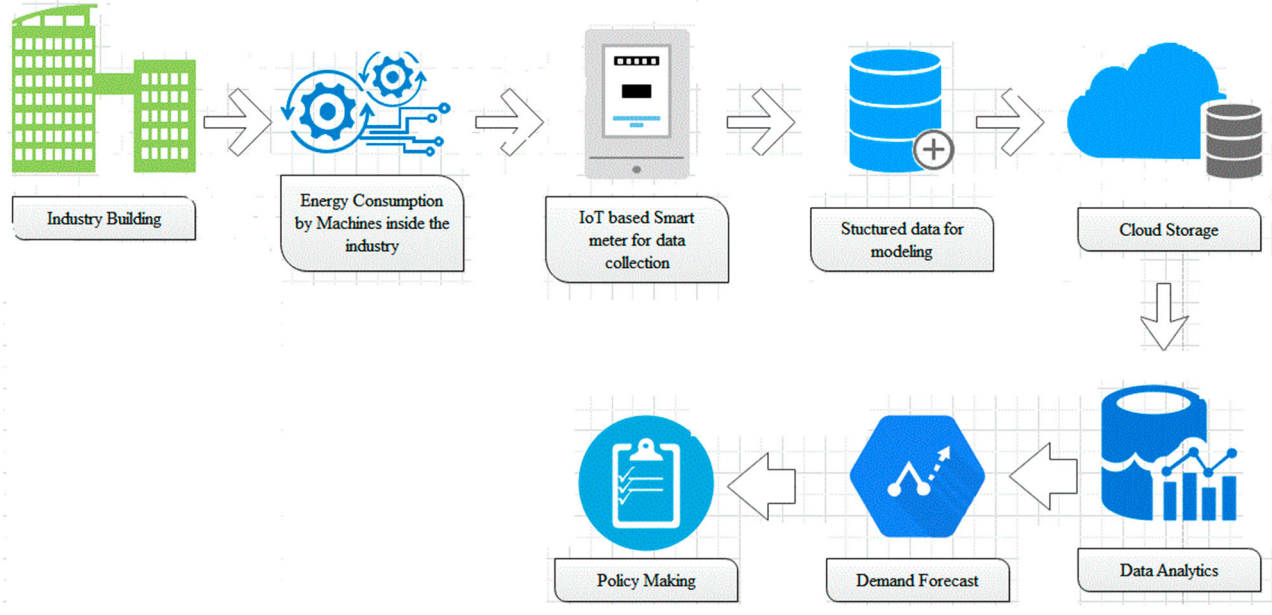


Figure 2. Overall flow of proposed system.

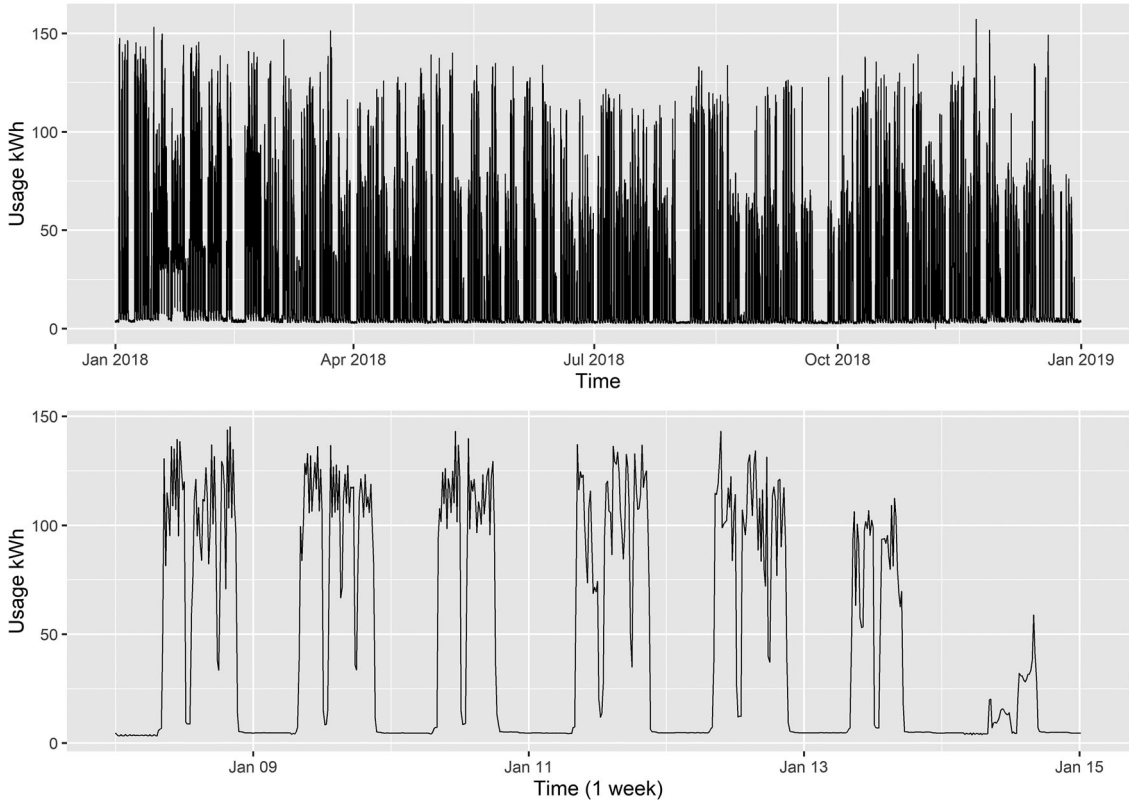


Figure 3. (Top) Steel industry energy consumption measurement for the whole study period. (Bottom) Steel industry energy consumption for the first week.

Most of the work pertaining to energy consumption used weather variables to forecast energy consumption based on weather variables impact (Candanedo et al., 2017). In this research, weather variables have no impact on energy consumption patterns because the steel industry considered in this study is in open space and has no heaters or cooling systems. Table 3 shows a full overview of the dataset. Additional characteristics are created by the date/time variable, such as the number of seconds from midnight for each day (NSM), the period or weekday position and the day of the week. This process is called Feature Engineering which involves deriving additional features

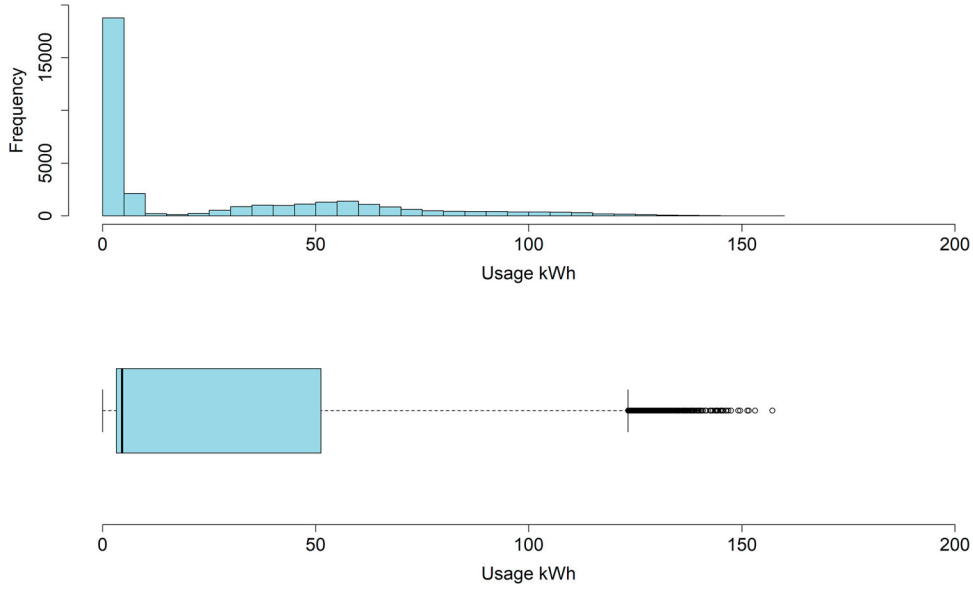


Figure 4. Distribution of energy consumption in the industry: histogram (top) and boxplot (bottom). The histogram indicates the frequency in the interval of the energy consumption. The boxplot shows a black line where the median is centred.

Table 3. Data variables and descriptions.

Data variables	Abbreviation	Type	Measurement
Industry energy consumption	Usage	Continuous	kWh
Lagging current reactive power	LagRP	Continuous	kVarh
Leading current reactive power	LeadRP	Continuous	kVarh
$tCO_2(CO_2)$	CO_2	Continuous	Ppm
Lagging current power factor	LagPF	Continuous	%
Leading current power factor	LeadPF	Continuous	%
Number of seconds from midnight	NSM	Continuous	S
Week status	Wstatus	Categorical	Weekend (0) or a Weekday (1)
Day of the week (Monday Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday)	Dweek	Categorical	Sunday, Monday … Saturday
Load Type	Ltype	Categorical	Light load, medium load, maximum load

from existing variables. This is done to improve the efficiency of the statistical learning algorithms.

Dataset and exploratory analysis

Using the create data partition function of Classification and Regression Training (CARET), the full one-year dataset was divided into training and test validation datasets. For training the models we used 75% of the data, with remaining 25% for testing purposes. Table 4 shows the dimensions of the testing and training sets. Figure 5 presents pairs plots displaying the links between all the energy consumption factors in the industry and was produced with the help of the psych package. The figure shows the bivariate scatter plots below the line the histogram plots along the axis, and above the line the Kendall correlation values are shown. Correlation of 1 is a full positive correlation, while −1 indicates a negative correlation and 0 signifies no correlation. Figure 4 shows a positive correlation between industrial energy use and tCO2 (0, 7.8). The carbon dioxide emissions begin with the industry's work initiation. The correlation elevates to 0.46 as the second major correlation, which is between Usage and LagRP. There is a positive correlation between LagPF and LeadPF. Usage and LeadRP show a negative correlation of −0.11. These values show that the considered independent variables are correlated with the dependent variable.

Table 4. Training and testing dataset.

Dataset	Number of observations
Training	26,281 and 10 variables
Testing	8759 and 10 variables

Data feature filtering and its importance

Data analysis is done to understand the data better. Hourly heat maps for five successive weeks were obtained to define any time patterns, and they are displayed in Figure 6. Figure 6 was constructed with the ggplot, gridExtra, and ggthemes (Arnold, 2017; Auguie, 2017; Wickham, 2016) packages. The energy consumption in the steel industry has a powerful time pattern. Energy consumption is lower during the weekend, compared to other days. Energy consumption begins to increase from 8 am and it stays steady until 8 pm. So the additional feature-engineered variables obtained make their contribution to the prediction algorithms, especially in the cases of tree-based and rule-based algorithms, because these additional variables (which are categorical) help in creating proper decision criteria or rules.

The dataset holds several parameters or characteristics that are appropriate for obtaining the crucial characteristics that enhance energy consumption prediction for the industry. The Boruta package was used to pick all appropriate factors (Kursa & Rudnicki, 2010). For several predictive algorithms the Boruta algorithm is used for variable filtering (Belgiu & Drăguț, 2016; Nguyen et al., 2015; Strempel et al., 2013). Two random variables, rv1 and rv2, are given to evaluate the Boruta Algorithm performance. Selecting a variable or a function from the Boruta decreases the complexity and interpretability of the model considerably.

The Boruta package relates the significance of the attributes to that of the shadow attributes by shuffling the original attributes. From Figure 6 it is evident that the Boruta algorithm have identified the two additional random variables rv1 and rv2, which do not involve in the prediction of energy utilization of industry. The two random variables inserted, rv1 and rv2, are between the shadow characteristics created by the Boruta algorithm (shown in blue): shadowMax, shadowMean, and shadowMin. In addition, the algorithm prioritizes the characteristics or variables in their order of importance, beginning with tCO2, to the variable of the week's least important day.

Figure 7 provides a lot of information about the importance of features for prediction but does not provide data on the effectiveness of the factors chosen for the Root mean square error (RMSE) values. For this purpose, Recursive Feature Elimination (RFE) is used for the identification of optimal variable count required to minimize RMSE (Fan et al., 2014). RMSE is used for evaluating an algorithm's efficiency in the regression process. This work uses RFE algorithm in the CARET package (Kuhn, 2015). In CARET, the RFE algorithm involves casting the variables of factor/category as dummy variables; these dummy variables are generated using the R package

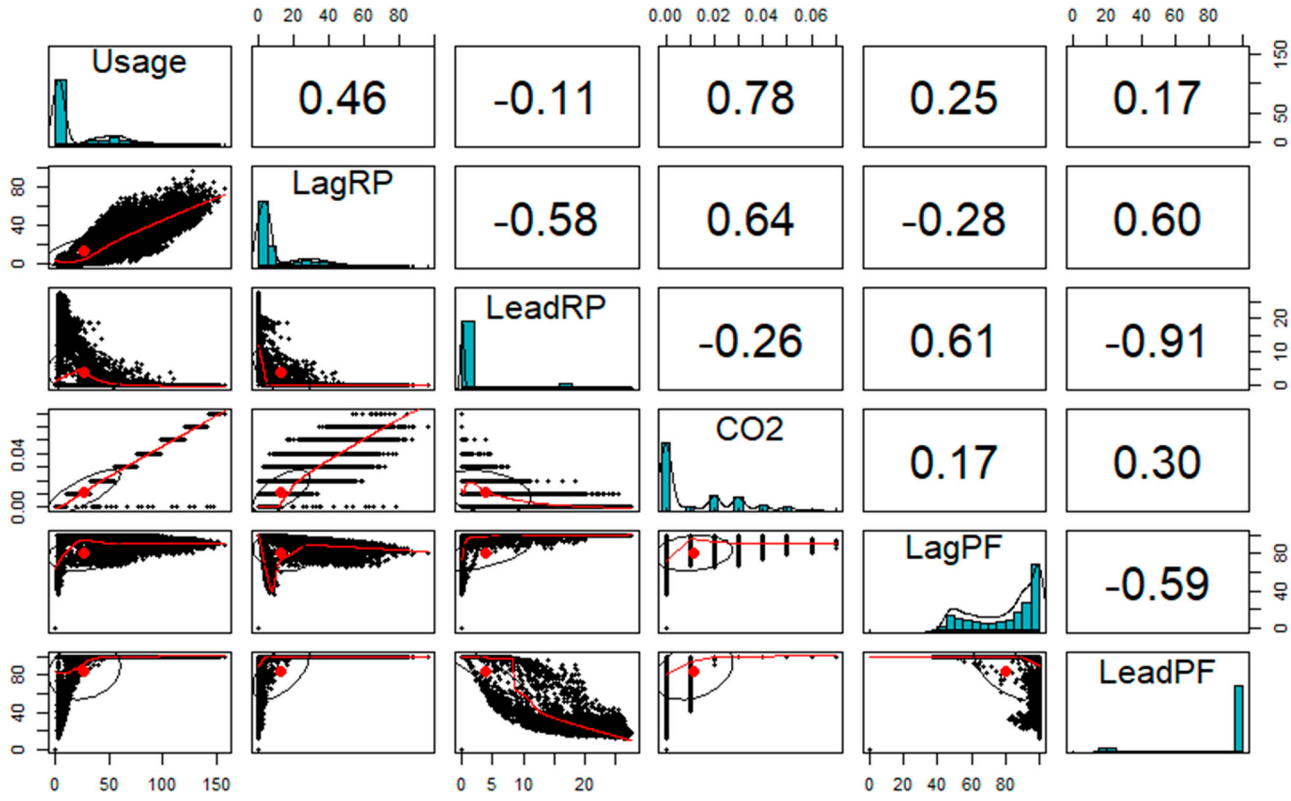


Figure 5. Pairs plots of the relationships between industry energy consumption, with LagRP, LeadRP, CO2, LagPF and LeadPF.

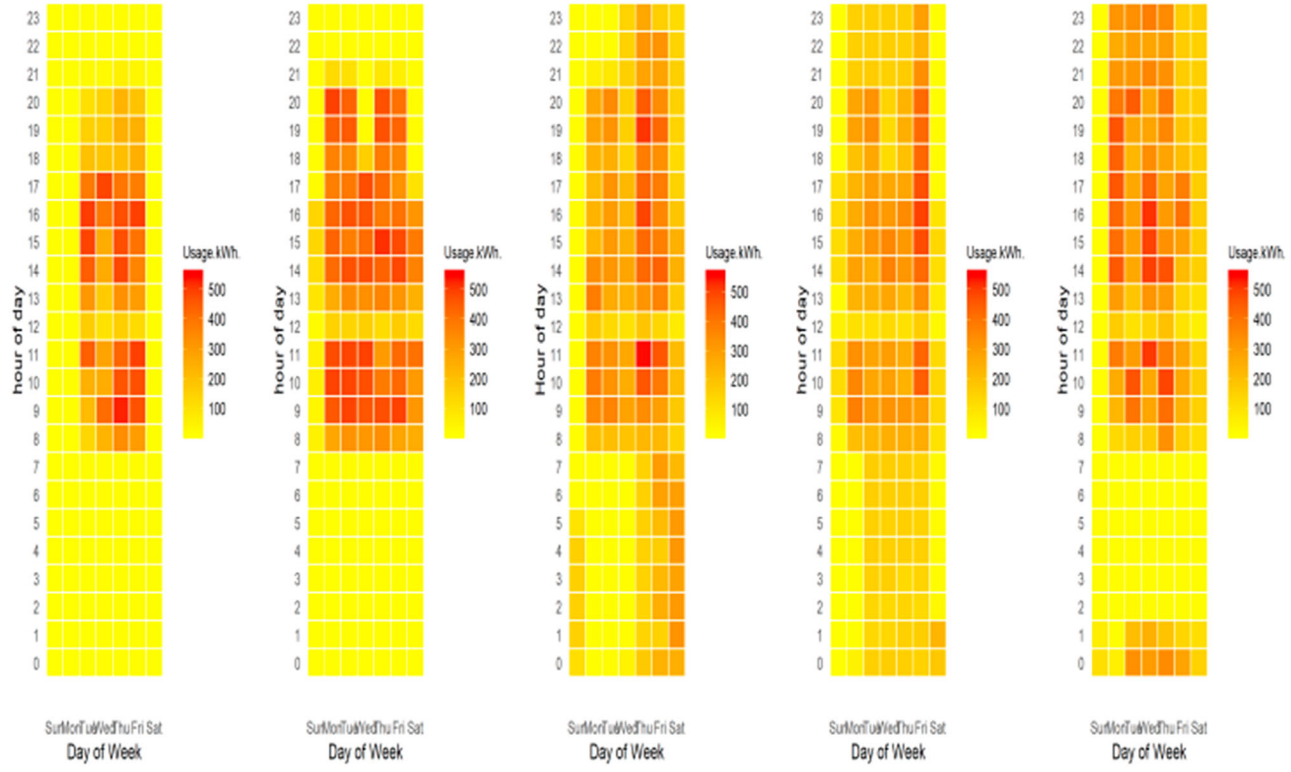


Figure 6. Hourly industry energy consumption heat maps for the first five weeks.

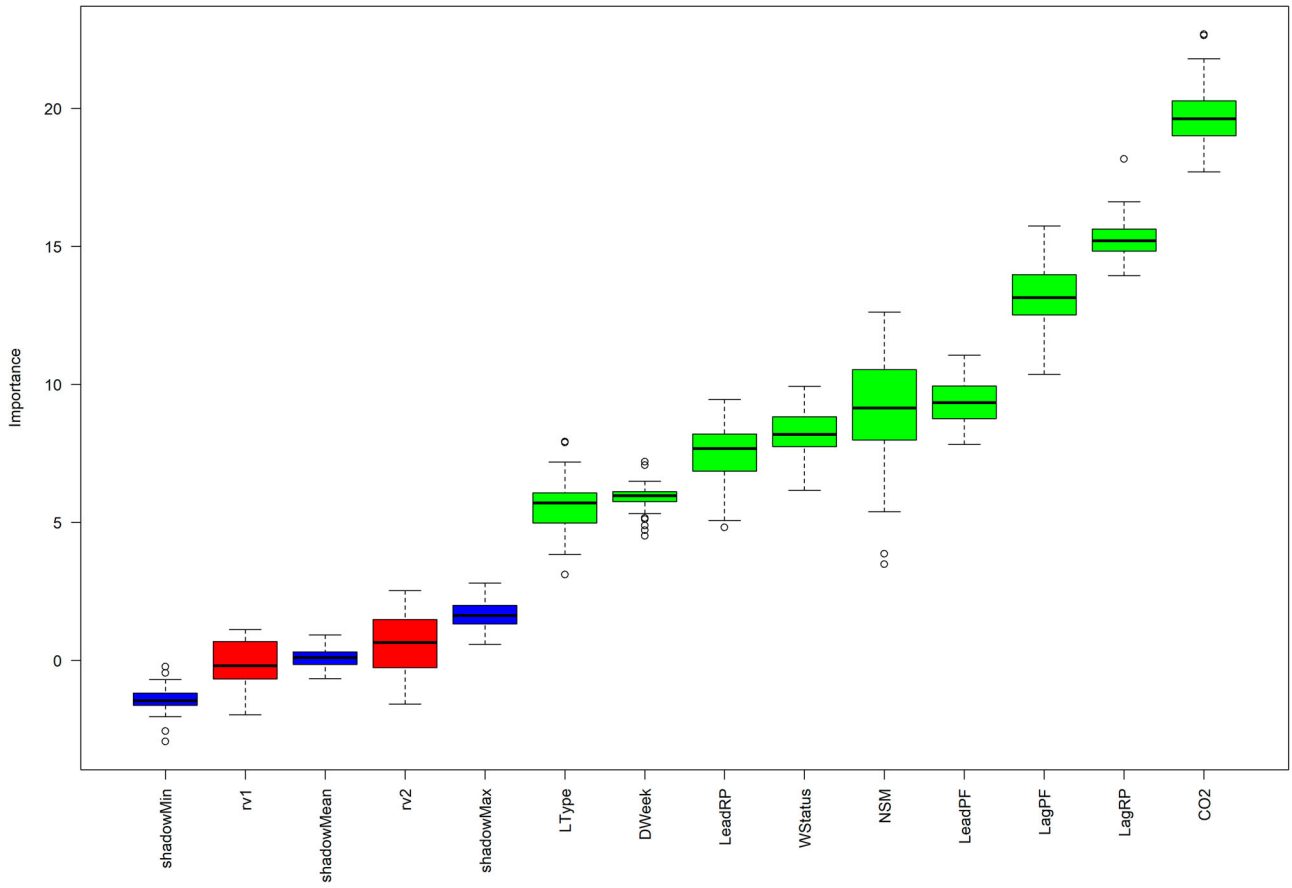


Figure 7. Variable importance and selection from the Boruta algorithm.

dummies (Brown, 2012). The factors; Dweek and Ltype are converted to dummy variables. Total number of predictors after that process is 18. The RFE algorithm utilizes the technique of random forest regression and was equipped with 10 pairs of cross-validation. Figure 8 displays the outcome of the RFE algorithm and the 18 predictors, which are considered to be optimal for the prediction is shown with dot fillings, Table 5 includes feature rankings from the RFE algorithm. For reducing RMSE, all the predictors are applicable. So, every predictor is used to evaluate the five regression models (GLR, KNN, CART, SVM, RF, and Cubist).

Evaluation indices

The prediction models were trained with a 10-fold cross-validation scheme to pick the finest tuning parameters. The doParallel set is used to accelerate calculations (Analytics & Weston, 2014). Multiple assessment criteria are used to compare regression model efficiency. RMSE, mean absolute error (MAE) and coefficient of variation (CV) are the performance evaluation indices used here.

RMSE is defined as the standard sample deviation between the known and expected values of the residuals. This value is used to determine major flaws, and to assess the variance response of the model. RMSE amplifies a mistake geometrically and helps to resolve major mistakes. Equation (6) defines RMSE:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{n}^{i=1} (y_i - \hat{y}_i)^2}{n}} \tag{6}$$

where y_i is the actual measurement (energy consumption), $\hat{y}_i$ is the predicted value from the regression algorithms, and n is the number of measurements.

Unlike RMSE, MAE monitors the proximity of the forecasts to the possible results. It calculates the average error magnitude between the forecast and the recorded values, disregarding over or under error flow. Formula (7) defines MAE:

$$\mathrm{MAE} = \frac{\sum_{n}^{i=1} |y_i - \hat{y}_i|}{n} \tag{7}$$

CV gets to decide how the overall mistakes in a forecast differ concerning the mean of the target. CV estimates the variation of a number sequence, independent of the measuring unit used for these figures. The precision is displayed as a percentage and is efficient for

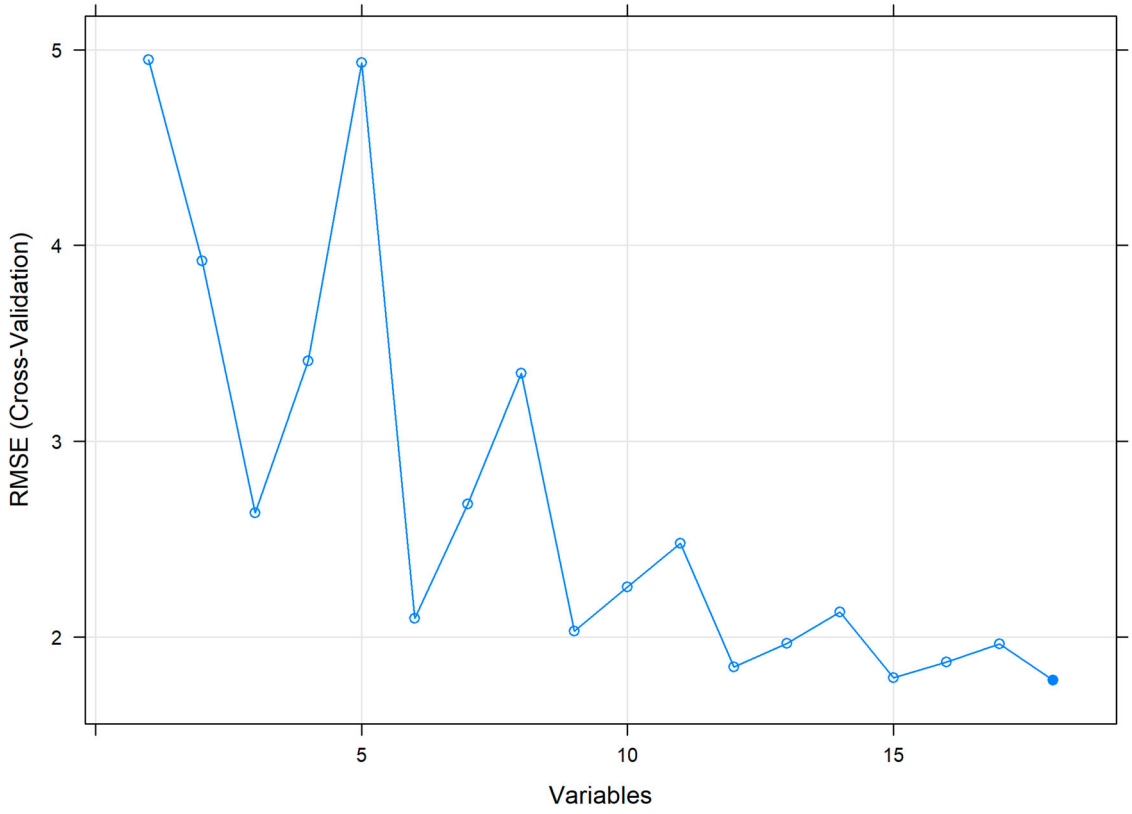


Figure 8. RMSE using the RFE algorithm.

assessing the output of the prediction model finding the error range. The mathematics expression for CV is Equation (8):

$$\mathrm{CV} = \frac{\sqrt{\frac{\sum_n^{i=1}(y_i - \hat{y})^2}{n}}}{\bar{y}_{\mathrm{data}}} \times 100 \qquad (8)$$

where $\bar{y}_{\mathrm{data}}$ is the average energy consumption.

All three matrices are combined to evaluate the models. The prediction model that provides lower values for RMSE, MAE, and CV is considered the best model.

Model development

After creating the dataset with additional features and identifying the important features via the Boruta and RFE algorithms, model development begins. When designing a prediction model, it is important to choose ideal hyperparameters to detect and handle the error values of the prediction algorithms. R language has a grid search function in CARET package to identify the finest parameters for a prediction model. GLR is a linear model and has no tuning parameters. A linear equation is used to fit the model with GLR. But in other regression algorithms, there exists a distinct tuning parameter for each model.

Table 5. Ranking of features using rfe algorithm.

1. tCO_2	2. LagRP	3. LagPF	4. LeadPF
5. NSM	6. Weekend	7. Weekday	8. Saturday
9. Sunday	10. LeadRP	11. Tuesday	12. Light Load
13. Maximum Load	14. Medium Load	15. Monday	16. Wednesday
17. Thursday	18. Friday		

CART is a decision tree and encloses the training data in the smallest feasible tree. The tuning parameter considered for CART is the complexity parameter (cp), which is the stopping criteria for CART models. If a higher value for cp is considered, the tree-building process becomes more complex, and leads to extensive computation time. So, it is important to find an optimum value of cp in order to save time and computation costs. A grid search for cp values ranged within the following: {0.0001, 0.001, 0.01, 0.1, 0.2, 0.3}. The optimal value for cp is found to be 0.2, and that was used to build the model. Figure 9 shows the grid search results for the CART model. In the figure, it can be seen, RMSE remains constant for 0.2 and 0.3, and so the optimal value was chosen as 0.2.

The SVM-radial basis kernel model needs two tuning parameters in addition to the predictors, namely sigma and cost, where cost controls the cost of misclassification, and sigma is the parameter of the radial basis kernel. The values 0.1, 0.35, and 0.4 were used for the sigma parameter, and the search for cost ranged within the

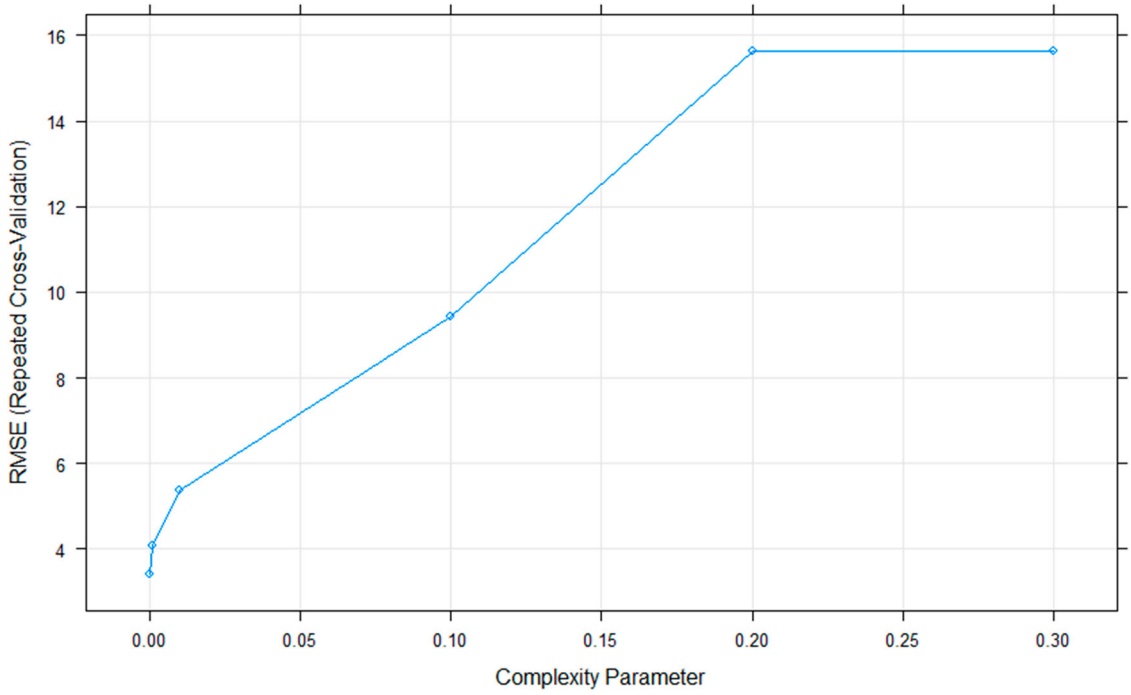


Figure 9. Grid search results of CART model.

following: {1,3,4,5,8,10,12,13}.The optimum values are achieved for the SVM tuning parameters sigma (0.1) and the cost (13) with the grid search shown in Figure 10.

KNN has a parameter for the number of neighbours, K, which needs optimization. Because the KNN model predicts values by averaging the neighbours, the number of neighbours needed is important in order to make the best prediction. The grid search for KNN ranged from 1 to 18, and the optimal value for K was found to be 2, as shown in Figure 11. When K is set at 2, the RMSE value is small.

RF is a model based on a tree (James et al., 2013). RF performance relies on multiple regression trees. However, a random sample of chosen predictors is used to

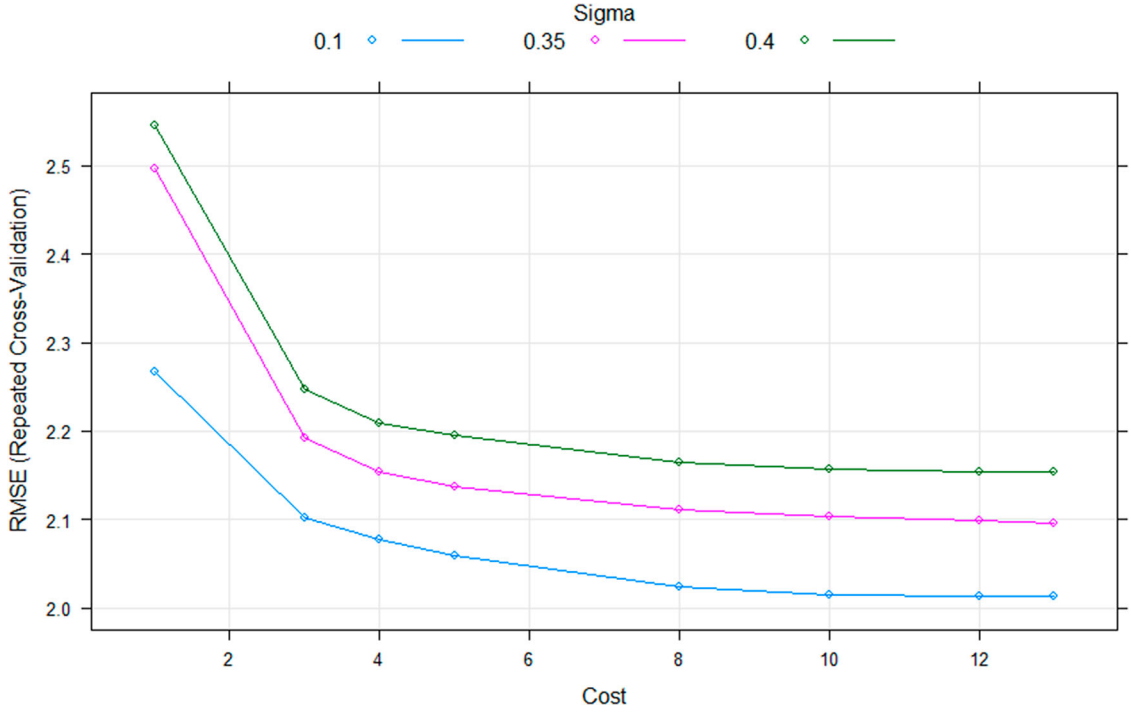


Figure 10. Grid search of SVM-radial basis kernel model.

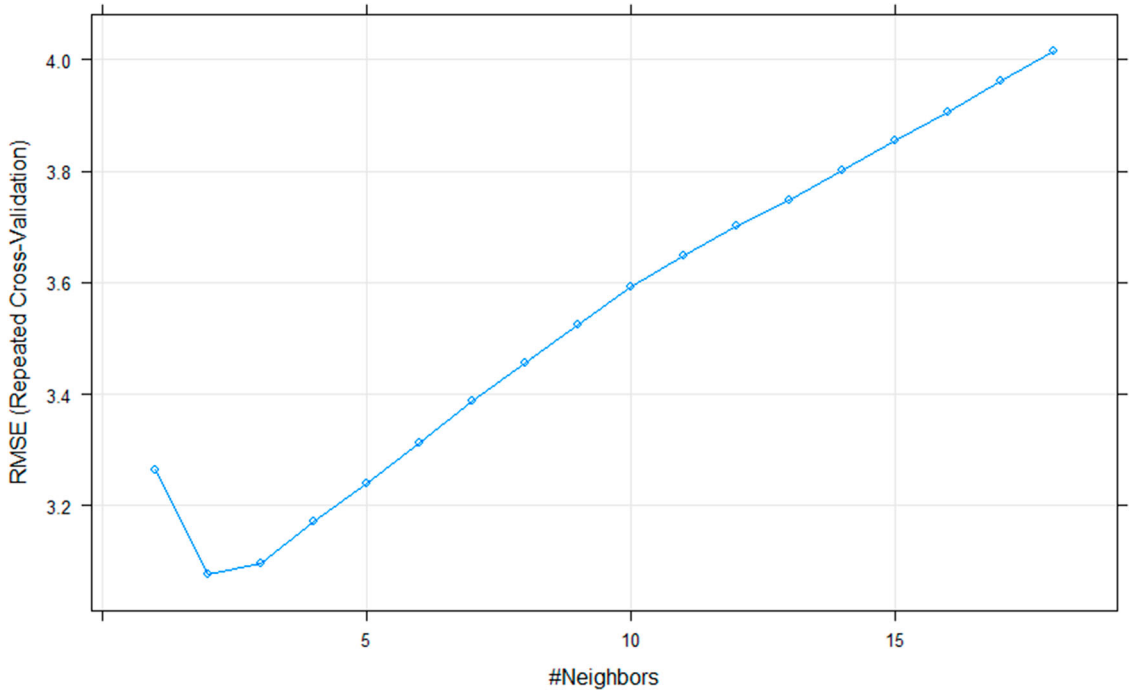


Figure 11. Grid search results of KNN model.

build each tree. This is accomplished through decorrelation of the trees and improvement of the forecast. RF needs to seek the best possible number of trees, but the number of predictors is chosen at random. From Figure 12, the optimum number of predictors is 18, which implies that RF takes all predictors into account, and that the RMSE value remains constant after 300 trees.

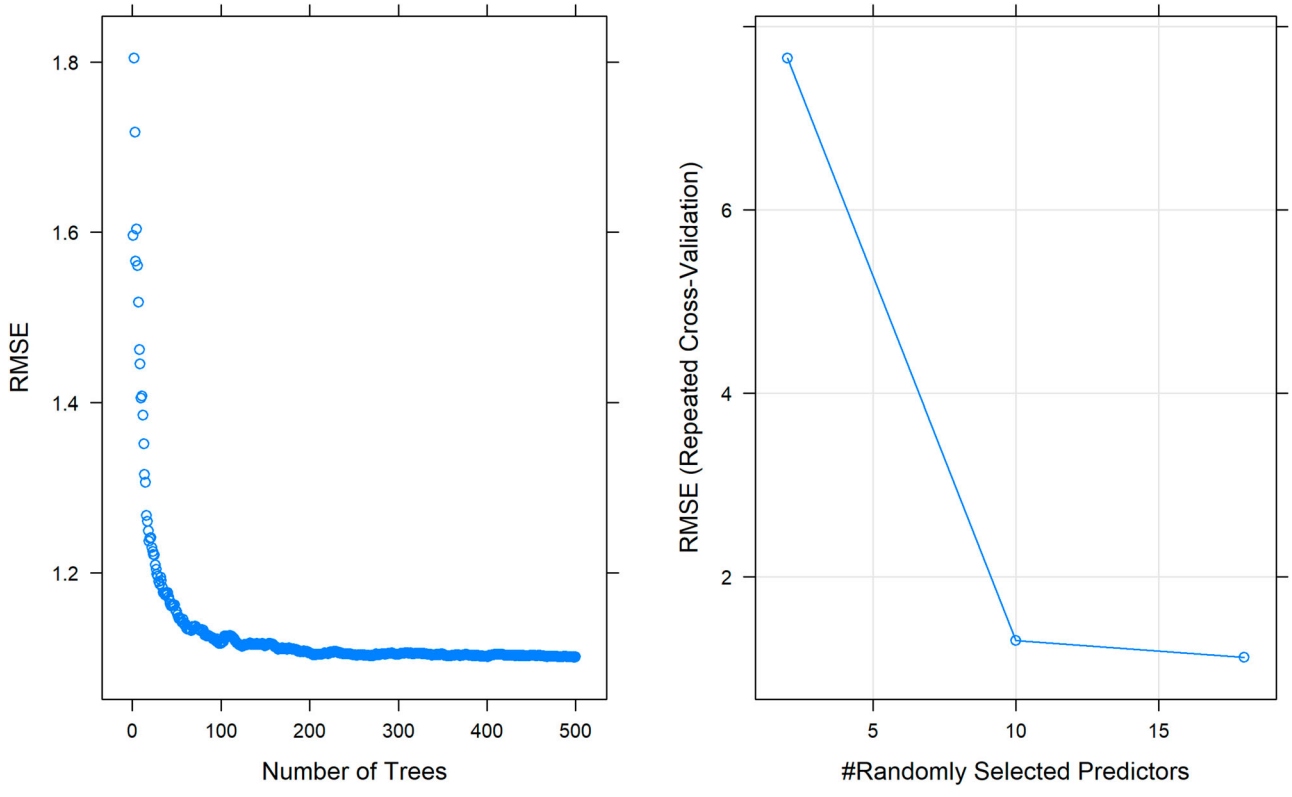


Figure 12. Grid Search results of RF.

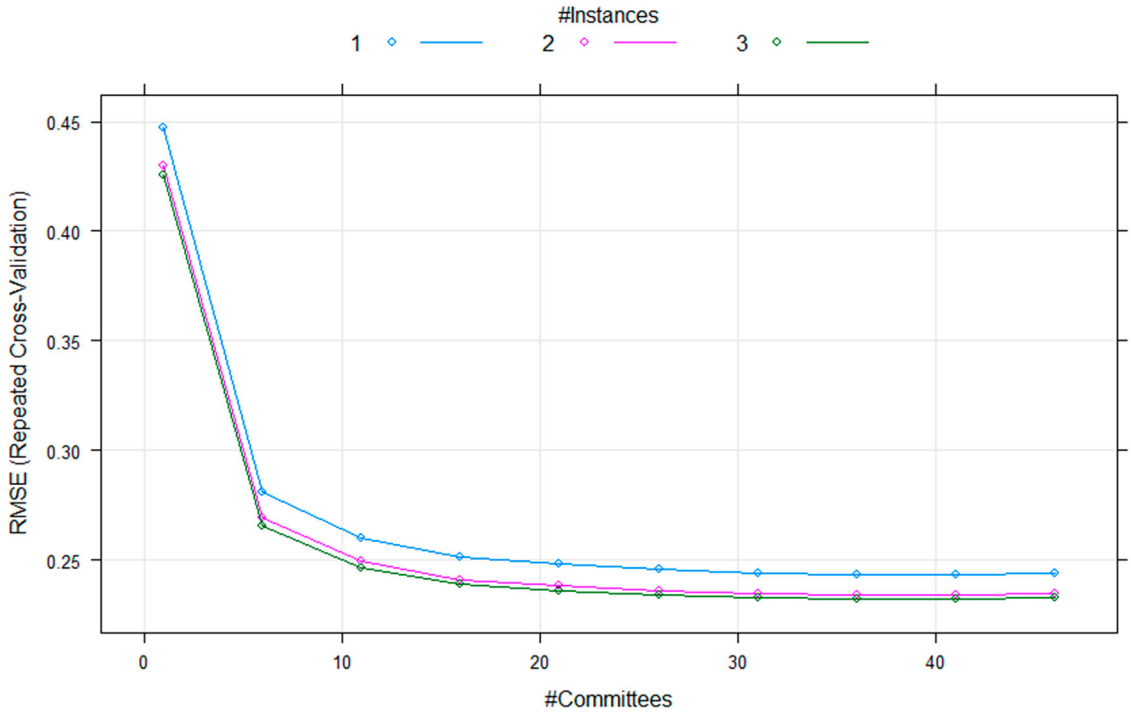


Figure 13. Grid search results of CUBIST model.

The Cubist regression model has two hyperparameters, which are to be tuned. (committees and neighbours). The grid search for the committees parameter ranged between 1 and 50, and the values 1, 2, and 3 were used for the neighbours parameter. The optimum value for committees is 36, and for neighbours, it is 3. The grid search result of the Cubist model is shown in Figure 12. From Figure 13, it can be seen that the RMSE value stays constant after the committees value of 36 and for neighbours at 3. So, the RMSE model becomes stable, and these values were chosen as optimal for developing Cubist model.

Results

Six regression-based models (GLR, CART, SVM, KNN, RF, and Cubist regression) were trained with their optimal tuning parameters using 10-fold cross-validation as discussed in Model development section. Each model performance is evaluated using the testing dataset, after training each model with the training set. Table 6 presents the performance of each of these models from training and testing sets. RMSE, MAE, and CV are the error measures used to determine the models mean error ranges and were summarized in Table 6. The optimal hyperparameters for each of the models were also summarized in Table 6.

Discussion

When compared to the CUBIST model, conventional models did not present a very high performance, which is very well evident in the results. CUBIST model produces a relatively high error values. This makes it a reliable method for the energy consumption prediction, which is primarily due to quality of data used. No pre-processing is done through our data and is free of missing values. Besides, it is necessary

Table 6. The models performance.

		Training			Testing		
Models	Optimal hyperparameters	RMSE	MAE	CV (%)	RMSE	MAE	CV (%)
GLR	*Intercept = TRUE*	4.6163	2.5387	16.8766	4.8586	2.5618	17.6764
CART	*Cp=0.2*	5.4005	2.9170	19.7434	5.6925	2.9569	20.7201
SVM	*Sigma = 0.1* *Cost = 13*	1.8930	1.5196	6.9205	1.9705	1.5438	7.1691
KNN	*K=2*	3.3078	1.4146	12.0930	3.6082	1.4826	13.1275
RF	*mtry = 18* *ntree = 300*	0.5072	0.1475	1.8542	1.1266	0.3601	4.0990
Cubist	*Committees = 36* *Neighbours = 3*	0.1122	0.0322	0.4104	0.2410	0.0547	0.8770

for regression models to learn from the data provided. So, Our dataset helped the models to learn with finely tuned regression models. For prediction, practitioners can go through tree-based models or rule-based methods. It is highly debatable to choose either of the case. Nevertheless, with the dataset type considered, the performance differs and necessitated further evaluation and comparison.

The model with lower values for RMSE, MAE, and CV is considered the best. As can be seen from Table 6, the Cubist model produces the lowest RMSE, MAE, and CV values in both training and testing. The performance of CART was worse, with higher error values. Ensemble strategies are used in CART to build the RF model. It is evident from the result that RF model produces better results than CART. This shows the impact of ensemble strategies used for building the RF model using CART. Next to the Cubist model, the performance of RF is considerably high. The performance of the SVM was comparatively better, when compared with KNN, CART, and GLR. The SVM model showed high stability in both training and testing. The RF model showed high performance instability in training and testing. The error values for RF in testing were almost three times higher than its performance during training.

The GLR and CART models produced the highest RMSE values of 4.8586 and 5.6925, respectively with the testing dataset. This shows the data is not linearly representable or can be represented using simple regression trees. Although the CART model was enhanced using RF, the performance was still lower than the Cubist model. This shows that regression tree strategies are not more effective in the problem of predicting energy consumption. Cubist model is capable of producing lowest RMSE value of 0.2410 in the testing set which is lower than other models in comparison. Also, MAE, and CV values were lower than the other algorithms in comparison. This shows the superiority of Cubist model in comparison with other conventional algorithms implemented in the energy consumption prediction domain.

From the results it is found that Cubist constantly produce best results than other methods. The flexibility of this model facilitates it to perform better irrespective of the known and unknown choice rules. In this study, rule-based Cubist model is used. It implies that the model learns complex attributes about energy consumption. But, when conceiving any regression model, the overfitting of the model can represent a real problem. Anyhow, the model overfitting can characterize a real problem when conceiving any regression model. So, it is vital to make sure that the model is fed with preprocessed data.

Conclusion

For smart cities, it is crucial to have a thorough understanding of patterns in Industrial energy consumption. The optimization process works at two level. The principal case is with the electricity organizations, which can enhance the evaluating strategies and furthermore can render customized packages to certain community types. The subsequent case is with the public organizations, which can optimize resources by getting certain hourly rates of discrimination offered by the primary power organizations in the nations. This work proposes a data analytic system for electricity company and public administrations. Inevitably, this benefits the community as a whole, since saving energy is critical for the decrease of effect on environmental change and advance sustainable development. In this context, a prediction method was proposed, which detects patterns from big data time series, as modern smart industry generates this data type through the increasingly common smart meters.

This analysis focused on predicting the energy consumption in the data analytic enabled smart steel industry by using variables like lagging and leading current reactive power, lagging and leading current power factor, carbon dioxide emissions, and load types. Feature engineering was done to extract all the additional variables. Exploratory analysis was carried out to better understand the patterns in the data. A detailed explanation for building the models with their optimal tuning parameters was shown. This study contributes in many ways to the already existing field of research. First, a new dataset was adopted for building the models, which includes variables such as lagging and leading current reactive power, and lagging and leading current power factor, which were never used in the existing research work for energy consumption prediction. Second, a rule-based model was used for energy consumption prediction. Third, an analysis of the empirical models is concentrated with the prediction of energy consumption in a smart industry. The findings of this paper show that Cubist regression, which is part of rule-based learning, is an advanced empirical modelling technique that can facilitate traditional learning algorithms to enhance prediction efficiency. These findings can provide researchers with a fresh alternative to predict energy consumption in an industrial sector. Further it helps them increase the library of empirical modelling algorithms focused on the prediction of Industrial energy consumption. And this paper can give company executives or government an effective and potential way in which the industrial energy problem can be well managed economically and environmentally. Therefore, we hope the suggested

method can help company managers or governments optimize energy use or their energy policies in the various fields related to industrial energy.

The suggested analytic method was conducted for only the steel industry, which is one of the study's primary limitations. Clearly, if it had covered several industries, we might have found significant information, which might differ from the results in this paper. Energy consumption of an industry can be explored in association with its products, the geometry of the buildings, the sort of equipment used, etc. Future research can include analysing individual equipment energy consumption with consideration for the equipment's operating duration, and developing an application, such as an alarm system, that operates when an unusual energy consumption pattern is encountered (Valouch, 2012).

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Trade, Industry & Energy (MOTIE) of the Republic of Korea [grant number 20172010000730].

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